

Denis Vinokurov

Ph.D. Candidate in Mathematics

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Education

2027 (<i>expected</i>)	Ph.D. in Mathematics • Advisors: I. Polterovich, M. Karpukhin	<i>Université de Montréal</i>
2024	M.Sc. in Mathematics • Diploma with honors • Thesis: “The first eigenvalue of the Laplacian on orientable surfaces” • Advisor: A. V. Penskoi	<i>Independent University of Moscow</i>
2022	B.Sc. in Mathematics • Diploma with honors • Thesis: “Geometry of harmonic maps of superminimal tori into spheres” • Advisor: A. V. Penskoi	<i>HSE University</i>

Research Interests

Geometric analysis, variational and optimization problems, and nonsmooth analysis, including spectral geometry, eigenvalue optimization, harmonic maps, and minimal submanifolds.

Honors and Awards

2026 – Present	Canada Graduate Research Scholarship – Doctoral (CGRS D)	<i>Natural Sciences and Engineering Research Council of Canada (NSERC)</i>
2026	Doctoral Research Scholarship awarded; declined in favor of CGRS D	<i>Fonds de recherche du Québec (FRQNT)</i>
2023 – 2025	ISM Graduate Scholarship	<i>Institut des Sciences Mathématiques (ISM)</i>

Publications and Preprints

- D. Vinokurov. “Eigenvalue optimization via a first-variation formula”. 2026. arXiv: [2606.31869 \[math.SP\]](#). Submitted
- D. Vinokurov. “Geometric bounds for Steklov and weighted Neumann eigenvalues on Euclidean domains”. 2026. arXiv: [2604.03418 \[math.SP\]](#). Submitted, revisions requested
- D. Vinokurov. “Maximizing higher eigenvalues in dimensions three and above”. 2025. arXiv: [2506.09328 \[math.SP\]](#). Submitted
- D. Vinokurov. “Eigenvalue optimization in higher dimensions and p -harmonic maps”. **Geom. Funct. Anal.** 36.4 (2026), pp. 1244–1293. DOI: [10.1007/s00039-026-00751-3](#)
- D. Vinokurov. “Conformal optimization of eigenvalues on surfaces with symmetries”. **J. Lond. Math. Soc.** 112.6 (2025), e70386. DOI: [10.1112/jlms.70386](#)
- M. Karpukhin and D. Vinokurov. “The first eigenvalue of the Laplacian on orientable surfaces”. **Math. Z.** 301.3 (2022), pp. 2733–2746. DOI: [10.1007/s00209-022-03009-4](#)

Conference Talks

<i>Feb 2026</i>	Workshop: <i>Emerging Horizons in Geometric Spectral Theory</i>	<i>Isaac Newton Institute</i>
<i>Dec 2025</i>	<i>Winter Meeting of the Canadian Mathematical Society</i>	<i>Toronto</i>
<i>Jun 2025</i>	Workshop: <i>Geometric Problems in Linear and Nonlinear PDEs</i>	<i>Isle of Skye, Scotland</i>
<i>Dec 2023</i>	<i>Winter Meeting of the Canadian Mathematical Society</i>	<i>Montréal</i>

Selected Seminar Talks

<i>Feb 2026</i>	<i>Two informal talks on Palais–Smale sequences in eigenvalue optimization</i>	<i>Isaac Newton Institute</i>
<i>Feb 2026</i>	<i>Functional Analysis and Noncommutative Geometry Seminar</i>	<i>HSE University (online)</i>
<i>Nov 2025</i>	<i>Spectral Geometry in the Clouds</i>	<i>Online</i>
<i>Oct 2025</i>	<i>CRM/Montreal/Quebec Analysis Seminar</i>	<i>Université de Montréal</i>
<i>Oct 2025</i>	<i>Analysis Seminar</i>	<i>Cornell University (online)</i>
<i>Oct 2025</i>	<i>Spectral Geometry Seminar</i>	<i>Independent University of Moscow (online)</i>
<i>Feb 2025</i>	<i>CIRGET Junior Seminar</i>	<i>UQAM</i>

Participation in Schools and Workshops

<i>May 2026</i>	<i>Conference on Classical Analysis in Memory of Paul Koosis</i>	<i>Université de Montréal</i>
<i>Jul 2025</i>	ISM Discovery School: <i>Interactions between Convex Geometry and Spectral Analysis</i>	<i>Université de Montréal</i>
<i>Jun 2023</i>	ISM Discovery School: <i>Geometry and Spectra of Random Hyperbolic Surfaces</i>	<i>Université de Montréal</i>
<i>May 2023</i>	Workshop: <i>Spectral Geometry and Applications</i>	<i>Université Laval</i>

Teaching Experience

<i>Fall 2025</i>	Teaching Assistant, Linear Algebra	<i>Université de Montréal</i>
<i>Winter 2025</i>	Teaching Assistant, Euclidean Geometry	<i>Université de Montréal</i>
<i>Fall 2024</i>	Teaching Assistant, Differential Geometry	<i>Université de Montréal</i>
<i>Winter 2022</i>	Teaching Assistant, Differential Geometry	<i>Independent University of Moscow</i>
<i>Fall 2021</i>	Teaching Assistant, Analysis on Manifolds	<i>HSE University</i>
<i>Fall 2020</i>	Teaching Assistant, Analysis on Manifolds	<i>HSE University</i>

Mentorship

<i>Summer 2026</i>	Joint supervision of a CRM–ISM undergraduate research project with Prof. Galia Dafni	<i>Université de Montréal</i>
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Refereeing

Geometric and Functional Analysis; Journal of the London Mathematical Society; Mathematische Zeitschrift.

Languages

Russian (native), English (fluent), French (intermediate).